

Pramo Jewels

Inventory Management Specification

Document ID: PJ-IMS-001
Version: 1.0
Status: Approved for Development

Field	Value
Project	Pramo Jewels Premium E-Commerce Platform
Purpose	Define inventory management processes, stock lifecycle and synchronization rules
Audience	Inventory Managers, Backend Developers, Admin Team, QA
Version	1.0

Inventory Modules

Module	Capabilities
SKU Management	Unique SKU generation and validation
Stock Tracking	Available, Reserved, Sold, Returned, Damaged
Reservations	Reserve stock during checkout
Reconciliation	Automatic updates after payment/cancellation
Alerts	Low-stock and out-of-stock notifications
Bulk Operations	CSV import/export and batch updates
Audit Logs	Track every inventory change
Future Ready	Multi-warehouse/location support

Inventory Lifecycle

Received → Available → Reserved → Sold → Shipped → Delivered → Returned (if applicable) or Damaged. Every transition is recorded in the audit log.

Stock Rules

Prevent overselling through database transactions and row locking. Automatically release reserved stock when checkout expires or payment fails.

Product Attributes

Track SKU, barcode/QR (future), metal type, purity, weight, size, stock quantity, reorder level and storage location.

Inventory Transactions

Support stock-in, stock-out, manual adjustments, returns, exchanges, cancellations and inventory corrections.

Alerts & Notifications

Notify administrators when stock reaches reorder thresholds, products go out of stock or unusual stock movements are detected.

Integration

Synchronize inventory with products, orders, payments, shipping, returns, analytics and notifications.

Performance

Use indexed queries, caching, optimistic/concurrent-safe updates and queue-based synchronization for high-volume operations.

Roles & Permissions

Inventory Managers maintain stock. Admins approve adjustments. Super Admins configure inventory policies.

Acceptance Criteria

Inventory always reflects actual availability, reservations prevent overselling, every change is traceable and synchronization remains consistent across all modules.